Day 7 Activity: ggplot2 internals with `storms`

Wednesday, May 27, 2026

Your Name Here

Setup

```r
library(tidyverse)
```

We'll use the `storms` dataset that comes built into the tidyverse (via `dplyr`). It contains 6-hourly observations of every named Atlantic storm since 1975, from NOAA's HURDAT2 reanalysis.

```r
glimpse(storms)
```

```
Rows: 20,778
Columns: 13
$ name                         <chr> "Amy", "Amy", "Amy", "Amy", "Amy", "Amy",~
$ year                         <dbl> 1975, 1975, 1975, 1975, 1975, 1975, 1975,~
$ month                        <dbl> 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6,~
$ day                          <int> 27, 27, 27, 27, 28, 28, 28, 28, 29, 29, 2~
$ hour                         <dbl> 0, 6, 12, 18, 0, 6, 12, 18, 0, 6, 12, 18,~
$ lat                          <dbl> 27.5, 28.5, 29.5, 30.5, 31.5, 32.4, 33.3,~
$ long                         <dbl> -79.0, -79.0, -79.0, -79.0, -78.8, -
78.7,~
$ status                       <fct> tropical depression, tropical depression,~
$ category                     <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ wind                         <int> 25, 25, 25, 25, 25, 25, 25, 30, 35, 40, 4~
$ pressure                     <int> 1013, 1013, 1013, 1013, 1012, 1012, 1011,~
$ tropicalstorm_force_diameter <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ hurricane_force_diameter     <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
```

Useful variables for today:

- `name`, `year`, `month`, `day`, `hour` — when the observation was taken
- `lat`, `long` — where the storm was
- `status` — `tropical depression`, `tropical storm`, `hurricane`, …
- `category` — Saffir-Simpson scale (1–5), `NA` for non-hurricanes
- `wind` — maximum sustained wind speed, knots
- `pressure` — central pressure, millibars (lower = stronger)

Part 1

Fill in the blanks in each chunk. Each one practices a concept from the slides.

1.1 — Mapping vs. setting

The chunk below colors every point the same shade of teal. **Without changing anything else,** modify the code so points are colored by `status` instead.

```
ggplot(storms, aes(x = wind, y = pressure)) +
  geom_point(color = "#1c7293", alpha = 0.4)
```

```
# Your fixed version:
```

In **one sentence**, what's the difference between putting `color = "#1c7293"` *inside* `aes()` versus *outside* it?

[Your answer here.]

1.2 — Local mappings

The plot below colors *both* the points *and* the smoothers by `status` (so you get one smoother per status). Modify it so **only the points are colored**, but the smoother is a single curve fit to all the data.

```
ggplot(storms, aes(x = wind, y = pressure, color = status)) +
  geom_point(alpha = 0.2) +
  geom_smooth()
```

```
# Your fixed version:
```

1.3 — The `group` aesthetic

Run this code first. It tries to draw the path of every storm in 2005, but it produces a single zigzagging line that connects unrelated storms together.

```
ggplot(storms |> filter(year == 2005), aes(x = long, y = lat)) +
  geom_path()
```

Add **one aesthetic mapping** so each storm gets its own path. (Hint: which variable identifies a storm?)

```
# Your fixed version:
```

1.4 — `after_stat()` for proportions

Make a bar chart showing the **proportion** (not count) of observations in each `status` category. You should not pre-compute the proportions with `dplyr` — let ggplot do it via `after_stat()`.

Hint: you'll need `group = 1`. The slide on `after_stat()` explains why.

```
# Your code:
```

1.5 — Position adjustments

Run the code below. It stacks hurricane counts by category, within each month.

```
hurricanes <- storms |> filter(status == "hurricane")

ggplot(hurricanes, aes(x = month, fill = factor(category))) +
  geom_bar()
```

Make **two** more versions:

(a) One where each bar is normalized so every month has the same height, and segments show the *proportion* of each category.

(b) One where the categories are placed *side by side* within each month, not stacked.

```
# (a) position = "fill"
```

```
# (b) position = "dodge"
```

In **one sentence**, which of the three versions (original stacked, fill, dodge) would you use to answer the question *"In which month is Category 5 most common as a share of all hurricanes?"*

[Your answer here.]

Part 2

Pick ONE of the three questions below. Build a plot that you think answers it well, then write 2–3 sentences explaining what you see.

You're encouraged to use:

- multiple geoms in layers
- local vs. global mappings
- a stat or position adjustment we covered today
- a thoughtful color or alpha choice

You do not have to use all of them.

Question A — Wind vs. pressure

It's well-known among meteorologists that storm wind speed and central pressure have a very tight relationship. Build a plot that **shows this relationship clearly** and lets us see how `status` (or `category`) fits in.

Suggestion: layered geoms with a smoother on top, or color by category and an alpha tuned for the overplotting.

Question B — When is hurricane season?

Build a plot that shows how **hurricane activity varies through the year**. You could look at counts by month, intensity by month, or how the distribution of categories shifts as the season progresses.

Suggestion: `geom_bar()` *works well here with a position adjustment, or possibly a faceted version of something simpler.*

Question C — One season's storm tracks

Pick a single year (say, 2005 — a famously bad year — or any other). Build a map of `lat` vs. `long` showing the **paths of every storm** that season. Highlight one or two notable storms.

Suggestion: this needs the `group` *aesthetic, and probably two* `geom_path()` *layers with different data.*

```
# Your plot:
```

Your 2–3 sentence interpretation:

[Write here.]

Code appendix

```
library(tidyverse)
glimpse(storms)
ggplot(storms, aes(x = wind, y = pressure)) +
  geom_point(color = "#1c7293", alpha = 0.4)
# Your fixed version:

ggplot(storms, aes(x = wind, y = pressure, color = status)) +
  geom_point(alpha = 0.2) +
  geom_smooth()
# Your fixed version:

ggplot(storms |> filter(year == 2005), aes(x = long, y = lat)) +
```

```r
  geom_path()
# Your fixed version:

# Your code:

hurricanes <- storms |> filter(status == "hurricane")

ggplot(hurricanes, aes(x = month, fill = factor(category))) +
  geom_bar()
# (a) position = "fill"

# (b) position = "dodge"

# Your plot:
```